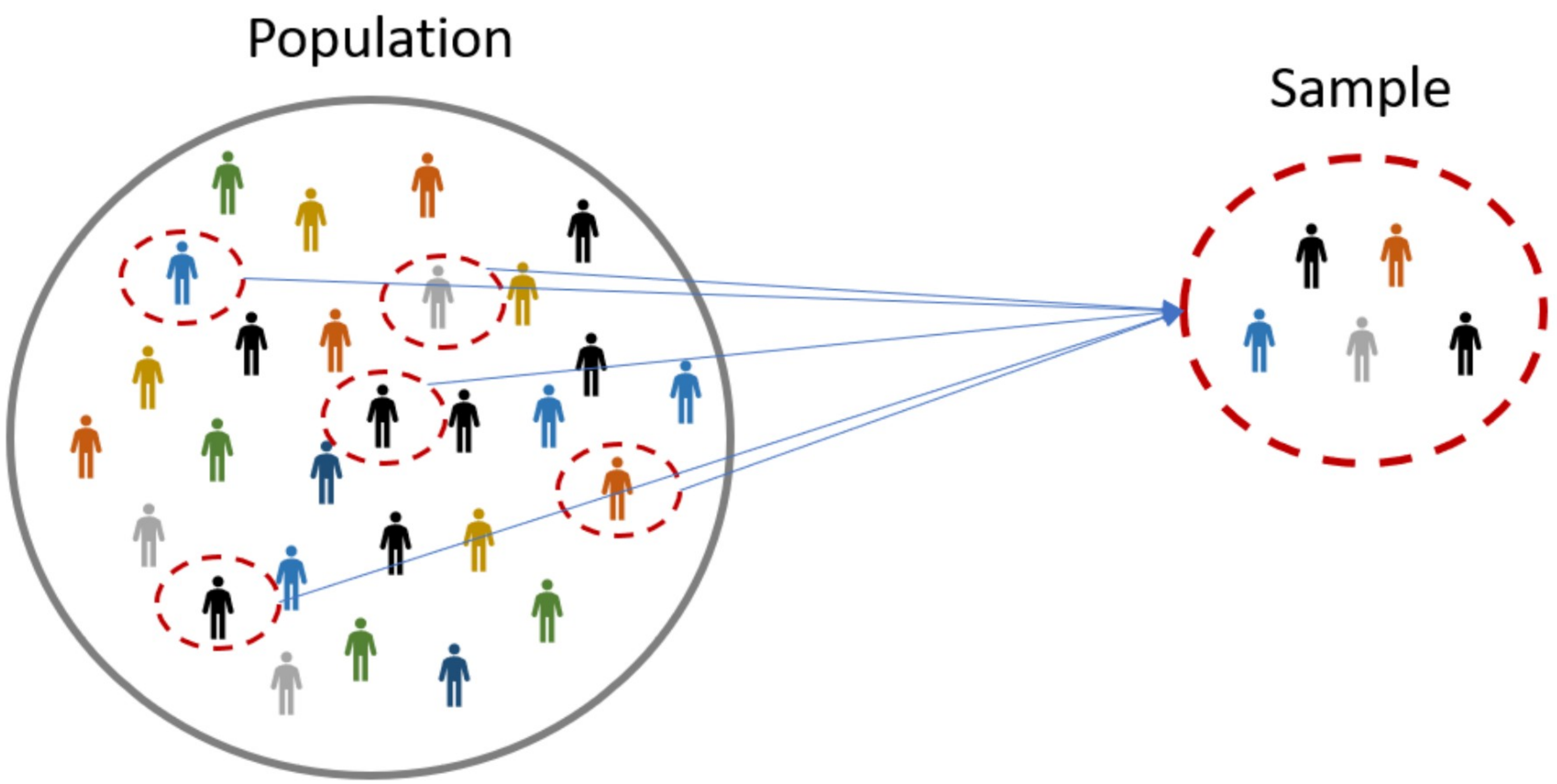
STAT 221- FUNDAMENTALS OF STATISTICAL DATA ANALYSIS

Week-12: Sampling and Statistical Inference-1

Shivakumar Jolad



SAMPLING

SAMPLING PROCESS

Population:
The group you want to generalize to
(e.g., professional workers around the world)

Sampling Frame:
A list from where you can draw your sample
(e.g., employees at 1-2 local companies)

The sampling frame must be representative of the population

Sample:
The actual units selected for observation
(e.g., a random selection of employees at each firm)

A sample is “a smaller (but hopefully representative) collection of units from a population used to determine truths about that population”

Why sample?

- Resources (time, money) and workload
- Gives results with known accuracy that can be calculated mathematically

Ch 8, Bhattacherjee, Anol, "Social Science Research: Principles, Methods, and Practices" (2012). http://scholarcommons.usf.edu/oa_textbooks/3

POPULATION AND SAMPLE

What is your population of interest?

To whom do you want to generalize your results?

- All doctors
- School children
- Indians
- Women aged 15-45 years
- Other

Can you sample the entire population?

When might you sample the entire population?

- When your population is very small
- When you have extensive resources
- When you don't expect a very high response

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Who do you want to generalize to?

The Theoretical Population

What population can you get access to?

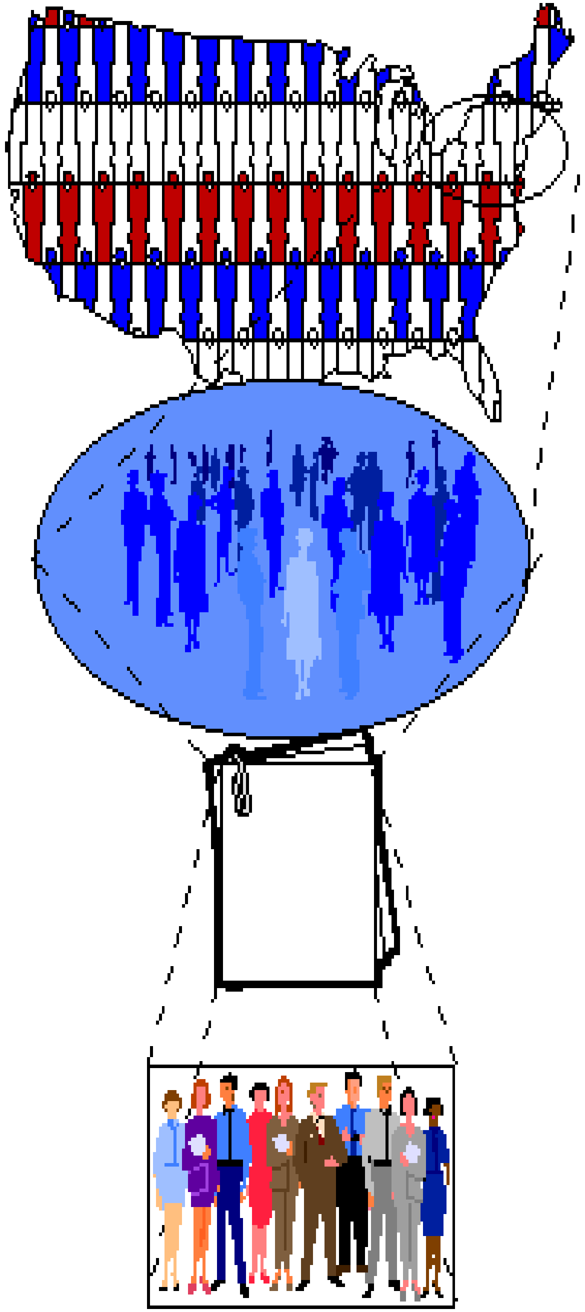
The Study Population

How can you get access to them?

The Sampling Frame

Who is in your study?

The Sample



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TYPES OF SAMPLES

Probability (Random) Samples

- **Simple random sample**
- **Systematic random sample**
- **Stratified random sample**
- Multistage sample
- **Multiphase sample**
- **Cluster sample**

Non-Probability Samples

- Convenience sample
- Purposive sample
- Quota

SAMPLING PROCESS

- Defining the population of concern
- **Specifying a sampling frame (a set of items or events possible to measure)**
- Specifying a sampling method
- Determining the sample size
- Implementing the sampling plan
- Sampling and data collecting
- Reviewing the sampling process

PROBABILITY SAMPLING

A **probability sampling** scheme is one in which every unit in the population has a chance (greater than zero) of being selected in the sample, and this probability can be accurately determined.

EPS: When every element in the population *does* have the same probability of selection, this is known as an **'equal probability of selection' (EPS) design**. Such designs are also referred to as 'self-weighting' because all sampled units are given the same weight**.**

Simple Random Sampling, Systematic Sampling, Stratified Random Sampling, Cluster Sampling, Multistage Sampling, Multiphase sampling

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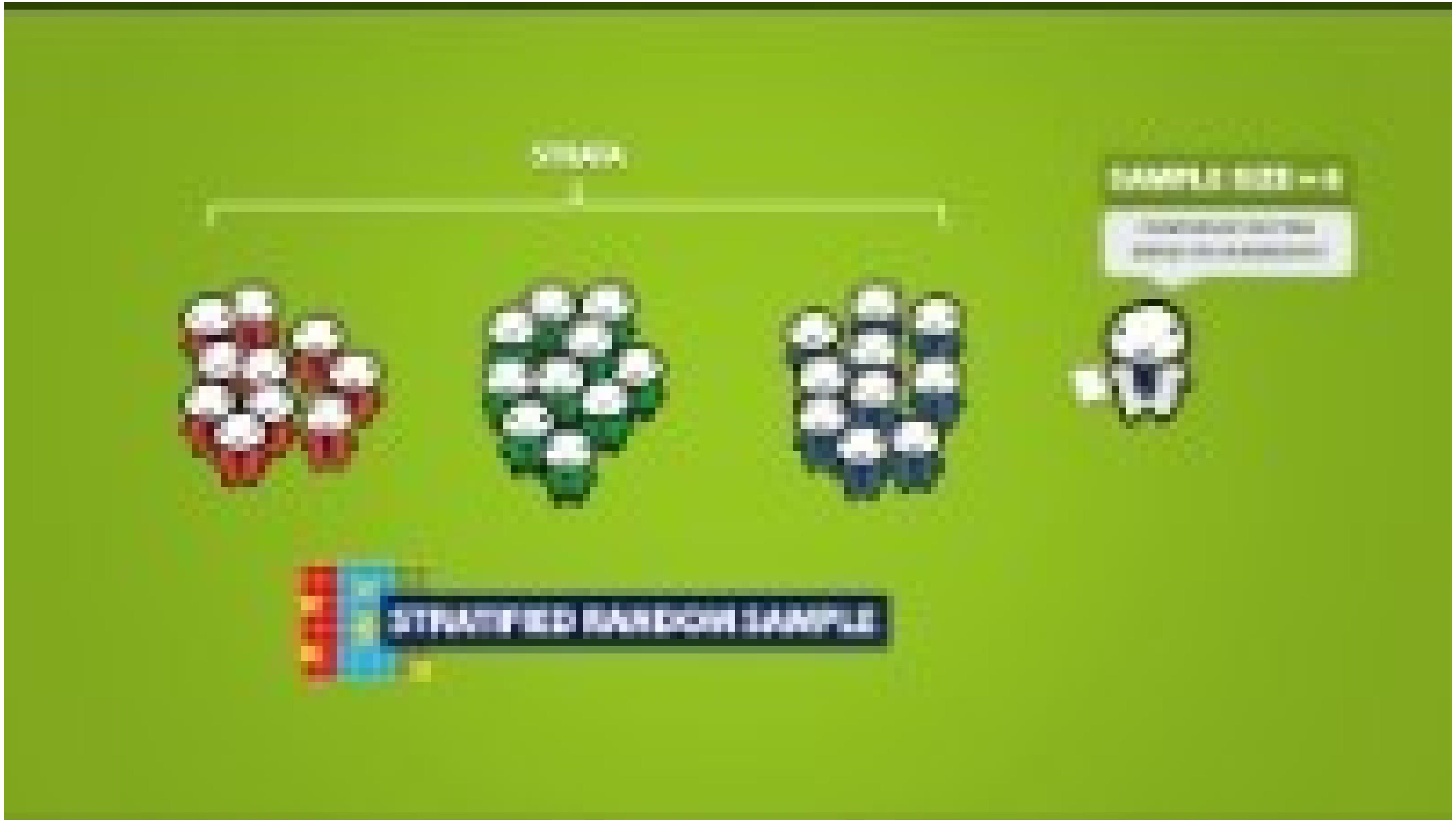
NON-PROBABILITY SAMPLING

Any sampling method where some elements of population have no chance of selection (these are sometimes referred to as 'out of coverage'/'undercovered'), or where the probability of selection can't be accurately determined.

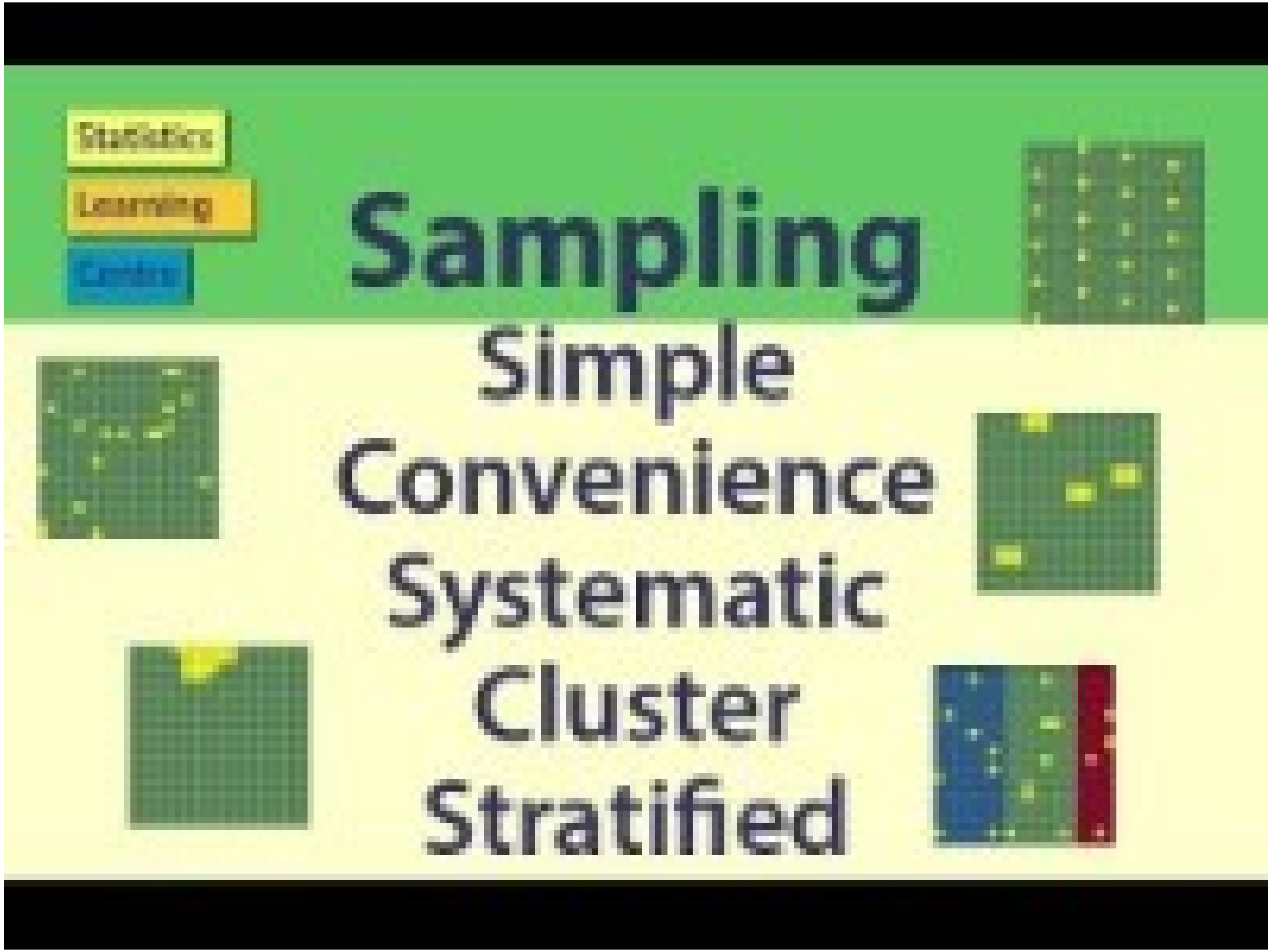
Because selection is non-random, nonprobability sampling does not allow the estimation of sampling errors, and may be subjected to a sampling bias. Therefore, information from a sample cannot be generalized back to the population.

Types: **Convenient, Quota, Purposive, Snowball**

From: Dr. Kanupriya



https://www.youtube.com/watch?v=pTuj57uXWlk



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Sampling: Simple Random, Convenience, systematic, cluster, stratified

SIMPLE RANDOM SAMPLING

- In this technique, all possible subsets of a population (more accurately, of a sampling frame) are given an equal probability of being selected.
- **The probability of selecting any set of *n* units out of a total of *N* units in a sampling frame is $^{N}C_{n}$.**
- Table of Random numbers (old time), Computer generated random numbers (most common)
- Sampling with Replacement (SWR)- and Sampling without Replacement (SWoR)

- Disadvantages :
  - **If sampling frame large, this method impracticable.**
  - **Minority subgroups of interest in population may not be present in sample in sufficient numbers for study.**

SYSTEMATIC SAMPLING

Systematic sampling relies on **arranging the target population according to some ordering scheme and then selecting elements at regular intervals through that ordered list.**

It is important that the starting point is not automatically the first in the list, but is instead randomly chosen from within the first to the kth element in the list.

Eg: **ASER Survey:** Every fifth household in a village

It is ***not 'simple random sampling'*** because different subsets of the same size have different selection probabilities - e.g. the set {4,14,24,...,994} has a one-in-ten probability of selection, but the set {4,13,24,34,...} has zero probability of selection.

STRATIFIED SAMPLING

Where population embraces a number of **distinct categories, the frame can be organized into separate "strata."** Each stratum is then sampled as an independent sub-population, out of which individual elements can be randomly selected.

Every unit in a stratum has same chance of being selected.

Using same sampling fraction for all strata ensures proportionate representation in the sample.

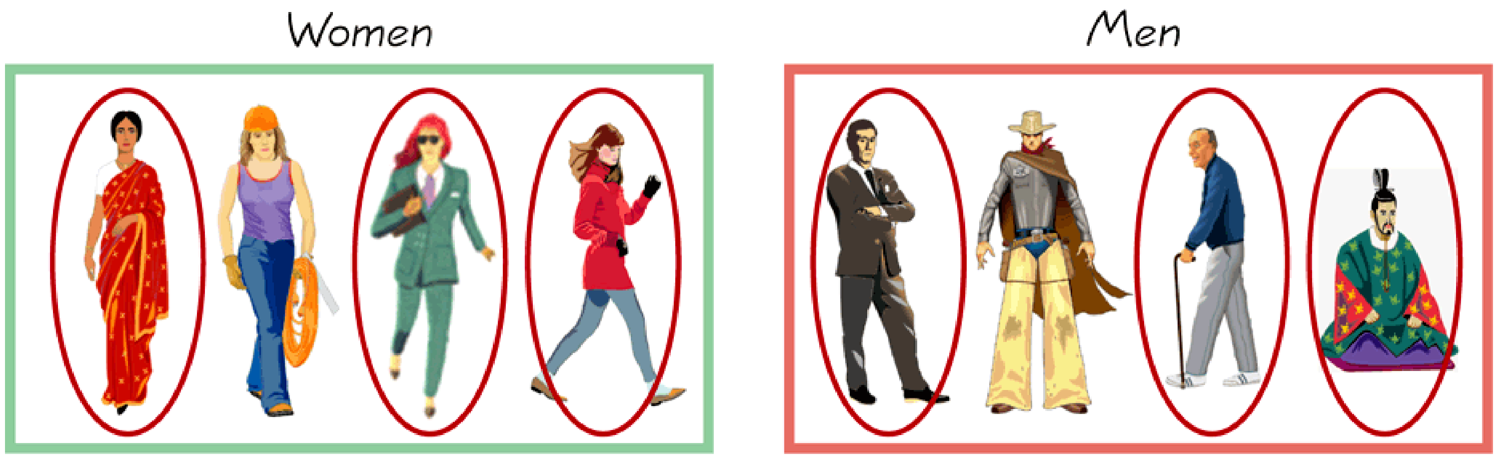
Adequate representation of minority subgroups of interest can be ensured by stratification & varying sampling fraction between strata as required.

Issue of Bias in sampling- **Non proportional stratification**

Proportional stratification

STRATIFIED SAMPLING

Draw a sample from each stratum



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CLUSTER SAMPLING

Population divided into clusters of homogeneous units, usually based on geographical contiguity.

Cluster sampling is an example of 'two-stage sampling' .

- First stage a sample of areas is chosen;
- Second stage a sample of respondents within those areas is selected.

Sampling units are groups rather than individuals.

- A sample of such clusters is then selected.
- All units from the selected clusters are studied.

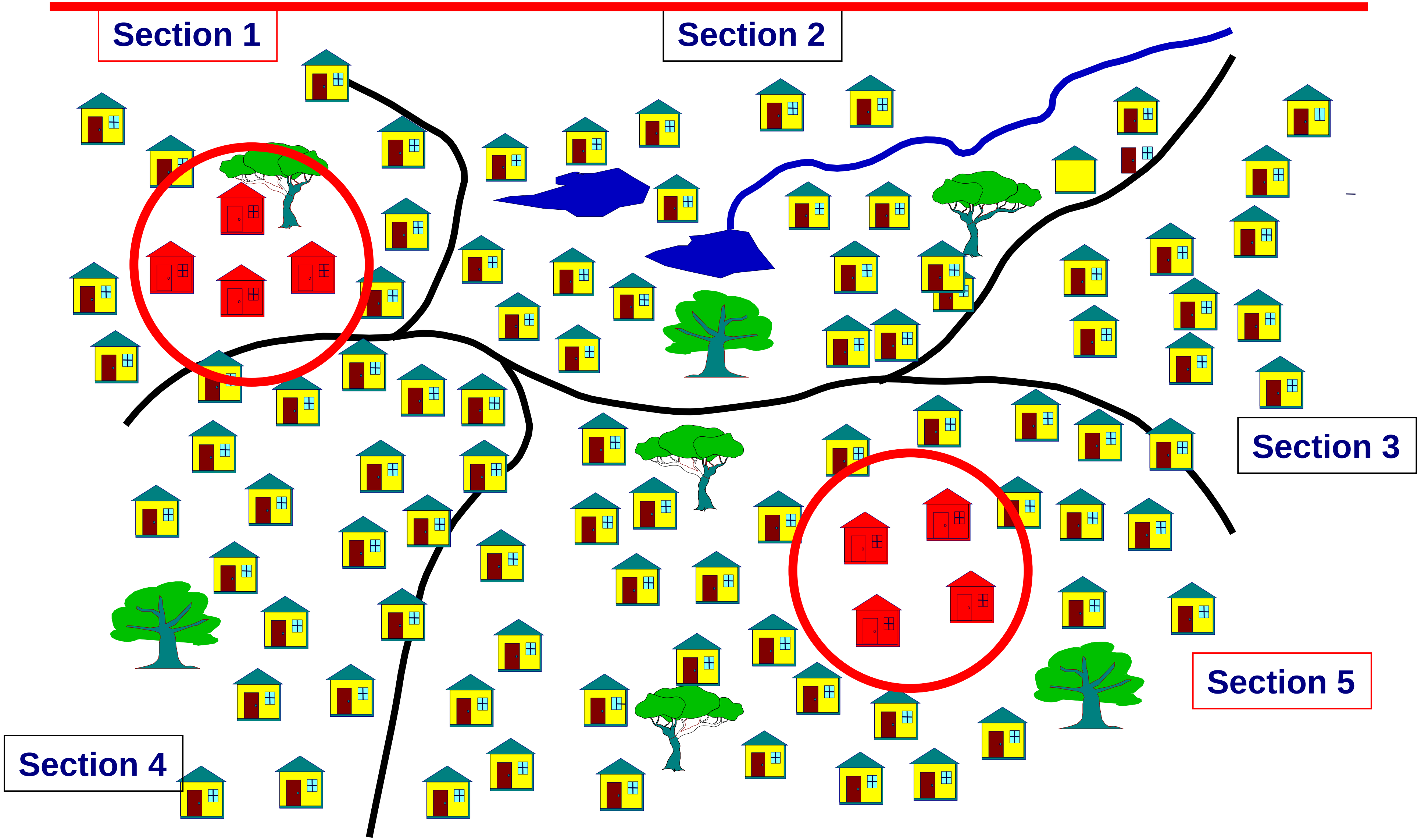
CLUSTER SAMPLING…….

Two types of cluster sampling methods.

One-stage sampling. All of the elements within selected clusters are included in the sample.

Two-stage sampling. A subset of elements within selected clusters are randomly selected for inclusion in the sample.

CLUSTER SAMPLING



MULTI-STAGE SAMPLING

Complex form of cluster sampling in which two or more levels of units are embedded one in the other.

- First stage, random number of districts chosen in all states.
- **Followed by random number of talukas, villages.**
- **Then third stage units will be houses.**

All ultimate units (houses, for instance) selected at last step are surveyed

NON-PROBABILITY SAMPLING: CONVENIENCE

Sometimes known as **grab or opportunity sampling or accidental or haphazard sampling.**

A type of nonprobability sampling which involves the sample being drawn from that part of the population which is close to hand. That is, readily available and convenient.

The researcher using such a sample **cannot scientifically make generalizations about the total population** from this sample because it would not be representative enough.

This type of sampling is most useful for pilot testing.

OTHER SAMPLING

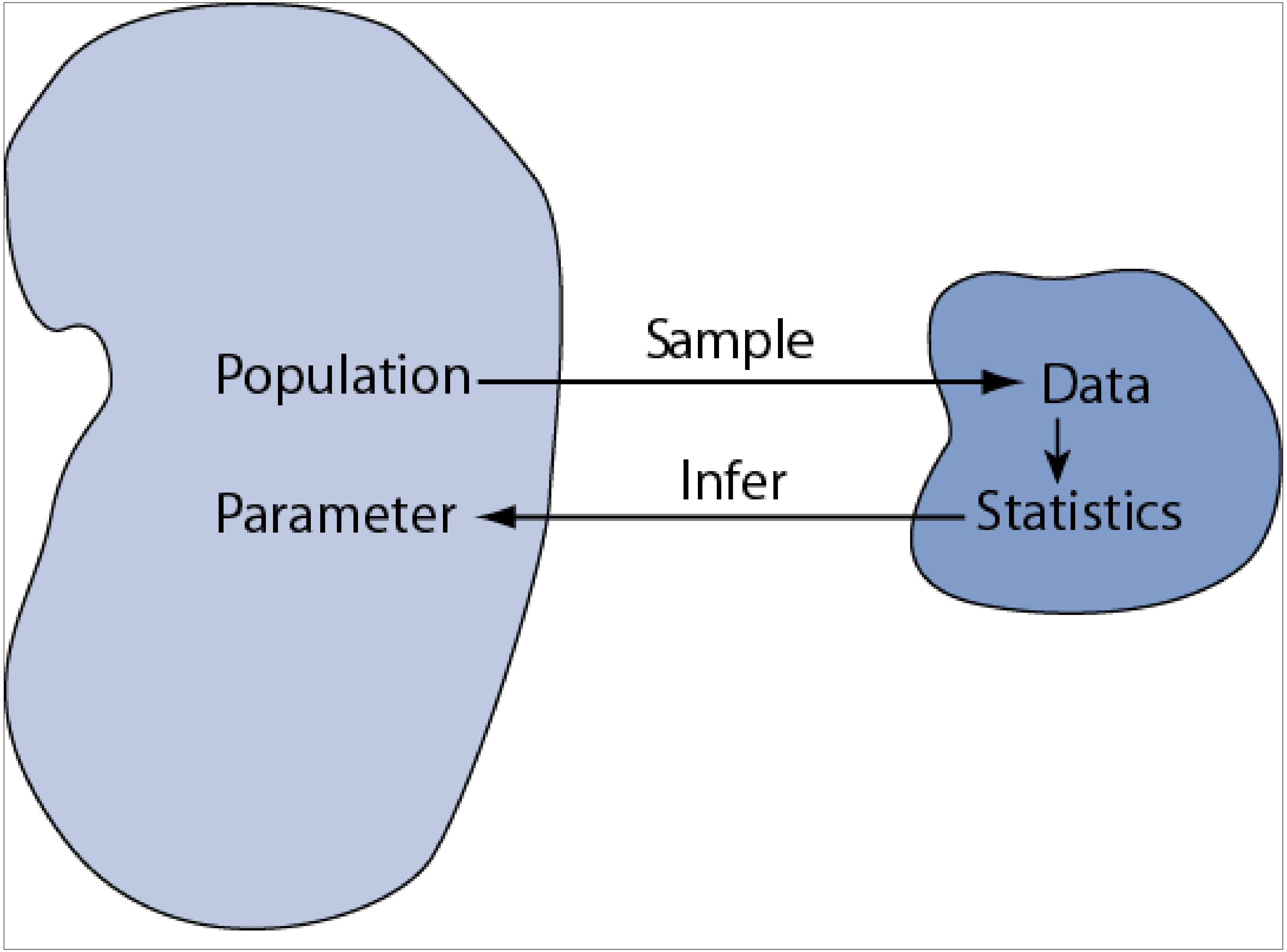
Judgmental sampling or **Purposive sampling**: The researcher chooses the sample based on who they think would be appropriate for the study. This is used primarily when there is a limited number of people that have expertise in the area being researched

Snowball Sampling: In snowball sampling, you start by identifying a few respondents that match the criteria for inclusion in your study, and then ask them to recommend others they know who also meet your selection criteria.

STATISTICAL INFERENCE

DISTINCTIONS BETWEEN PARAMETERS AND STATISTICS

	Parameters	Statistics
Source	Population	Sample
Notation	Greek (e.g., μ)	Roman (e.g., *xbar*)
Vary	No	Yes
Calculated	No	Yes

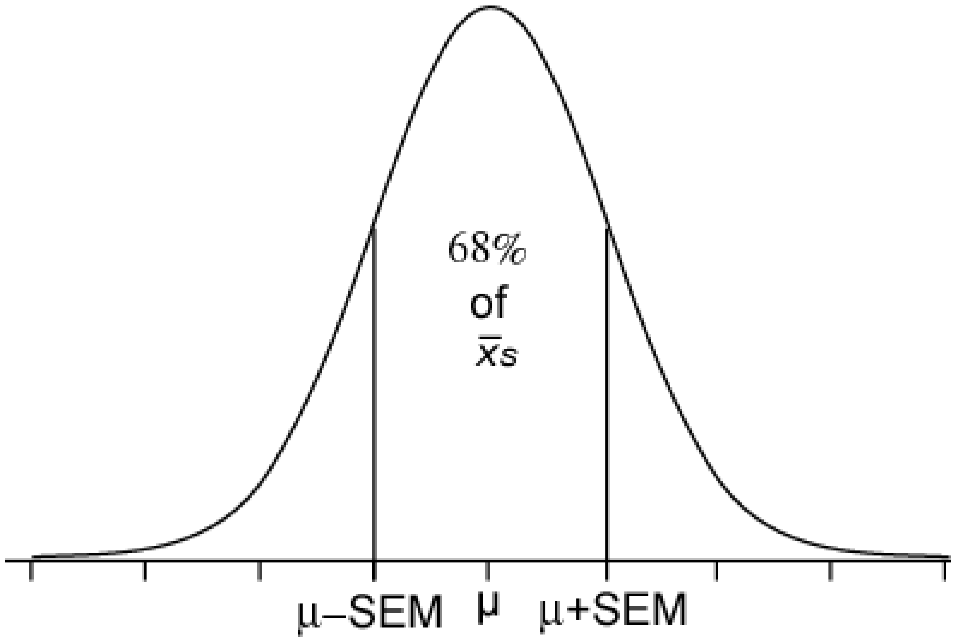

Sample
Population
Data
Infer
Parameter
Statistics

SAMPLING DISTRIBUTIONS OF A MEAN (INTRODUCED IN CH 8)

The **sampling distributions of a mean (SDM)** describes the behavior of a sampling mean

$$\bar{x} \sim N\left(\mu, SE_{\bar{x}}\right)$$

$$\text{where } SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$



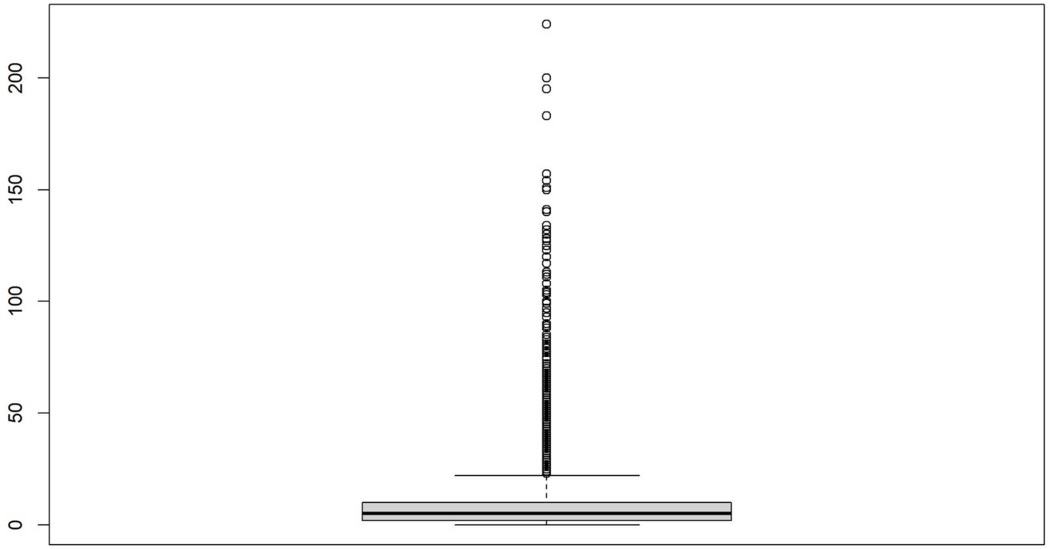
PUNE SCHOOLS EXAMPLE

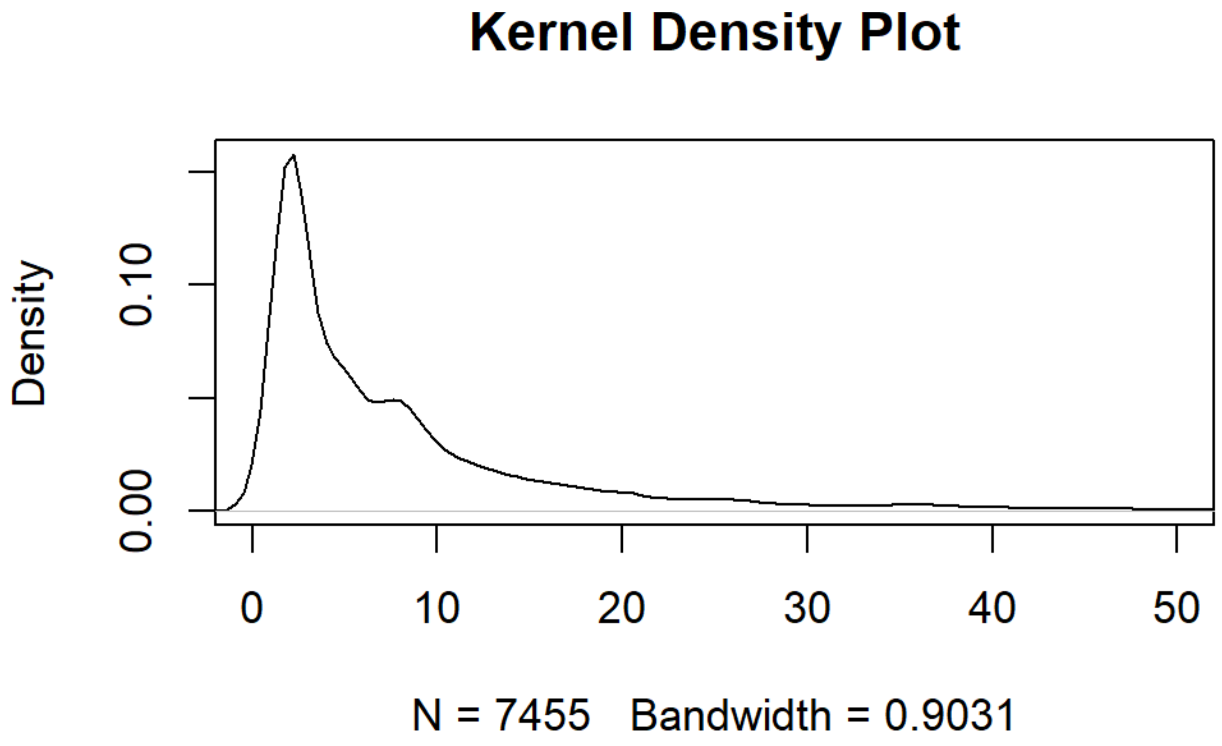
7455 schools

School Management	Dept of Edu	Tribal Welfare	Local body	Govt Aided	Private Unaided	Unrecognized	Social Welfare	KVS	JNV	Madrassa	Unknown	Total
1			2906	189	318	20	1					3434
2	2	2	1188	202	539	13	1			1		1948
3		5	5	279	205	2		14			1	511
4					4							4
5			4	45	47				1			97
6		13	14	370	346	1	1	1				746
7			34	220	189	1	4					448
8			13		61	4						78
10					16							16
11				41	132							173
Grand Total	2	20	4164	1346	1857	41	7	15	1	1	1	7455

TEACHERS IN SCHOOLS- ALL UDISE CENSUS

Mean – mu = 9.47
Sigma= 14.19





SAMPLE OF SCHOOLS AND TEACHERS

xbar= 9.45826
SE=1.412487

mu= 9.474581
sigma/
sqrt(num_schools_per_sample)
=1.420

Sample size =100
No of samples = 1000